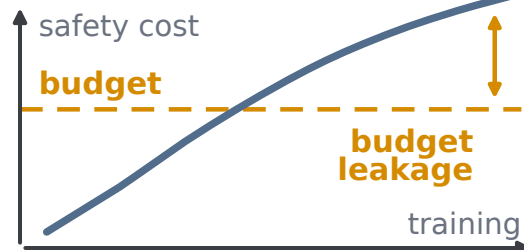


Penalty-based safe RL

$$\max_{\pi} J_r(\pi) - \lambda^\top \mathbf{J}_c(\pi)$$



Typical failure mode

One penalty vector prices all channels;
a rare channel gets traded away.

SafeTransport

one capacitated γ -flow bridge

1. Encode

2. Compose

3. Solve once

c_1 **Bridge impact**
hard $S^{(1)}$

c_2 **Liquid spill**
soft $S^{(2)}$

c_3 **Contact force**
hard $S^{(3)}$

c_K **Fire / tipping**
hard $S^{(K)}$

Hard minimum

$$S_{ss'}^{(k)} = \frac{b_k \omega_{ss'}^{(k)}}{\bar{c}_k(s)}$$

$$S = \min_k S^{(k)}$$

certificate

smooth Compose_{T_c}

$S^{(1)}$

$S^{(2)}$

$S^{(K)}$

**capacitated
 γ -flow Sinkhorn**

$$F_\varepsilon^* = \arg \min_{F \in \mathcal{F}_S}$$

$$\langle D, F \rangle - \varepsilon H(F)$$

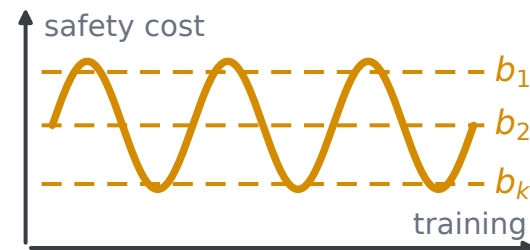
feasible target flow



$$F_\varepsilon^* \leq S$$

Constraint-based safe RL

$$\max_{\pi} J_r(\pi) \quad \text{s.t.} \quad J_{c_k}(\pi) \leq b_k \quad \forall k$$



Typical failure mode

Coupled dual updates oscillate
near several active budgets.

Conceptual positioning: penalties $\rightarrow$ one SafeTransport capacity solve $\rightarrow$ projections

PPO / SAC

PPO-Lag / FOCOPS

SafeTransport

one inner capacity projection

CPO

CUP / hard projection

unconstrained or penalty

external safety controller

local constrained step

constraint-dominant